WEEK 2 · DAY 8 · INTERN DISCUSSION 4

Video / Signal Quality Audit

1. SCOPE OF THIS AUDIT

Day 7 already confirmed file integrity (no corruption, frame counts match headers) and found the active transmission window within each video. Day 8 goes further: is the camera/transmitter setup actually static, is the transmitter reliably distinguishable from background, and does ambient lighting drift during capture? All three matter directly for how forgiving (or not) the ROI-detection and thresholding logic in Days 9-11 needs to be.

2. CAMERA & TRANSMITTER MOTION

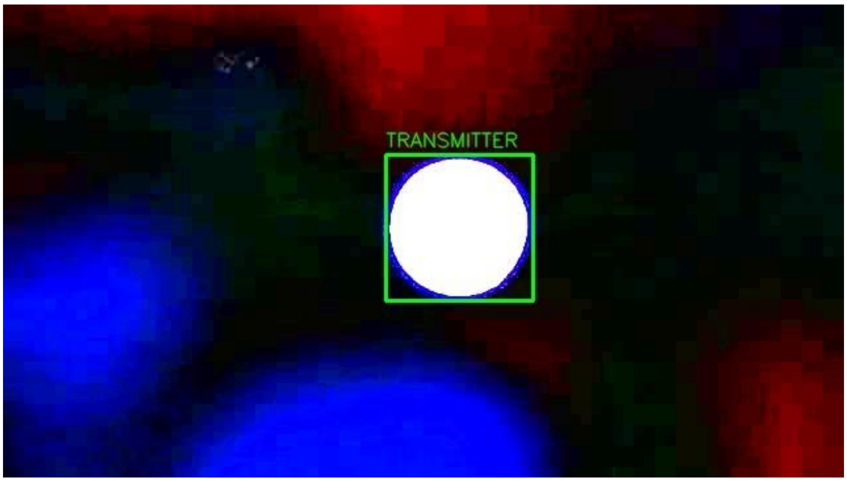
Method: at 6 evenly spaced checkpoints across each video's active window, the transmitter LED(s) were located by connected-component analysis (filtering out small non-transmitter bright spots by area), then matched across checkpoints to measure positional drift in pixels.

File	Max Positional Drift	Interpretation
1LED_92bps.mp4	0.31 px	Effectively static
2LEDs_92bps.mp4	0.33 px	Effectively static
3LEDs_92bps.mp4	2.08 px	Effectively static
4LEDs_92bps.mp4	4.01 px	Effectively static

All four videos show sub-5-pixel drift across their full active window — well within sensor/compression noise. Camera motion is not a meaningful risk for this dataset; a fixed ROI defined once per video (rather than per-frame re-detection) should be safe to use in Day 9-11 preprocessing.

3. TRANSMITTER VISIBILITY — AN EXTRACTION RISK WORTH FLAGGING

While measuring motion in the 1-LED video, a second, smaller bright region was found intermittently appearing in the upper-left of frame — separate from the actual transmitter. It is roughly 3-8% of the transmitter's pixel area and not collocated with it.



This matters because a naive "find the brightest point/blob in frame" ROI detector would occasionally lock onto this artifact instead of the real transmitter, specifically during frames where the true LED is OFF (OOK low state) and the artifact happens to be the brightest thing in frame at that instant. The fix used here — filter candidate blobs by a minimum area before treating anything as "the transmitter", resolved it, and is now a concrete requirement for the Day 9-10 ROI-detection code, not just a one-off note.

The source of the artifact itself (reflection, secondary indicator LED, sensor hot pixel cluster) hasn't been determined and doesn't need to be — the extraction logic just needs to be robust to its presence.

4. AMBIENT LIGHT VARIATION

Method: mean intensity of a fixed background patch (frame corner, away from the transmitter in all 4 videos), sampled at 25 points spread across each video's full duration — independent of the transmitter's own on/off state.

File	Background Mean	Std Dev	Range
1LED_92bps.mp4	2.45	0.86	1.42 – 5.34
2LEDs_92bps.mp4	1.93	0.96	1.06 – 4.28
3LEDs_92bps.mp4	1.45	0.12	1.27 – 1.72
4LEDs_92bps.mp4	1.39	0.21	0.98 – 1.89

Background levels are low and stable across all four files (all values near the sensor noise floor on a 0-255 scale, standard deviation under 1 in every case) — consistent with a dark, controlled capture environment rather than a room with variable ambient lighting. Ambient-light drift is not a meaningful risk for this dataset. The background colour-cast difference between videos noted on Day 7 (green/red/blue tint) is a separate, white-balance-level issue, not an intensity drift one, and remains worth asking the mentor about.

5. RAW DATA PRESERVATION

All analysis in this audit reads directly from the untouched files in raw/ — no video was re-encoded, trimmed, or overwritten. Brightness/gain adjustments used for visual inspection exist only as separate output images, never applied back to source files.

6. VIDEO-QUALITY / RISK TABLE

File	Readability	Motion	Ambient Drift	Visibility Risk
1LED_92bps.mp4	Clean	None	None	Artifact present — filtered
2LEDs_92bps.mp4	Clean	None	None	None found
3LEDs_92bps.mp4	Clean	None	None	None found
4LEDs_92bps.mp4	Clean	None	None	None found

7. EXTRACTION RISK CHECKLIST — FOR DAY 9-11 PREPROCESSING

- ROI detection must filter candidate bright regions by a minimum pixel area (~2000px for 1/2/4-LED videos, ~900px for the 3-LED video) to avoid locking onto the non-transmitter artifact found in Section 3.
- A fixed ROI per video, defined once from a confirmed fully-lit frame, is justified — sub-5px drift means per frame re-detection is not necessary and would only add noise.

- Trim or explicitly mask the dead-time lead/trail windows identified on Day 7 before running any extraction — e.g. don't process 0-3.5s of the 2-LED video.
- Do not assume every bright blob in frame is a transmitter — always cross-check candidate count against the expected N for that file (1/2/3/4) before accepting a detection.
- Ambient-light compensation is not needed for this dataset — background is stable — but the white-balance/colour-cast difference between files should still be normalized (e.g. convert to grayscale before ROI/intensity work, which already sidesteps this) rather than assumed irrelevant.

8. NEXT STEP

Day 9's ROI/signal-plan review can now propose a concrete, justified ROI-detection strategy ,fixed box per video, area-filtered blob detection, dead-time trimming, backed by the measurements in this audit rather than assumption.